

TIM 172B, Lecture # 13, MOT-II, 2/17/26

### Agenda

- Safety Inventory condition
- Demand statistics
- Determination of Safety Inventory, SS, and the Re-order point (ROP): Process
- Product availability condition (PAC)
- Probability Theory
- Safety Inv. based on PAC & Prob. Theory

# 1. Safety Inventory condition :

If the actual demand during the supplier lead-time,  $D_L$ , is less than the available supply,  $ROP$ , when the order is placed, then the product is available to all customers during the lead-time,  $L$ .

In symbols

If  $D_L \leq ROP$  , then the product (supply) is available to all potential customers

$\downarrow$                        $\downarrow$   
 actual demand          available supply  
 during the lead-time,  $L$

## Safety Inventory problem :

Determine:

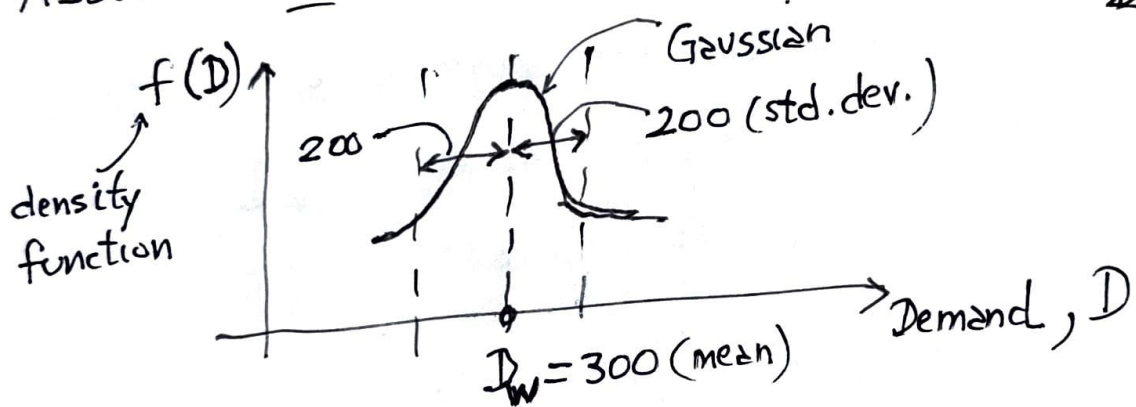
- Safety stock,  $SS$ , to meet a desired cycle service level,  $(CSL)_{desired}$
- $ROP$ , re-order pt.

Given : Demand statistics

(i) Average (mean) wweekly demand,  $D_w = 300 \frac{\text{units}}{\text{week}}$

(ii) standard deviation of wweekly demand,  $\sigma_w = 200 \frac{\text{units}}{\text{week}}$

Assume a normal or Gaussian distribution:



Coefficient of variation,  $cv = \frac{\text{std. dev.}}{\text{mean}} = \frac{200}{300} = 0.67$

(iii) supplier lead time,  $L = 2$  weeks

(iv) Desired cycle service level,  $(CSL)_{\text{desired}} = 95\%$   
 $\equiv 0.95$

for now,  $CSL =$  probability of the product being available to the customer during the supplier lead-time,  $L$

- (V) Continuous Review Policy : continuously monitor the inventory level, and when the inventory level drops to ROP, place a re-order 13.4

Process & Implementation (to determine ss & ROP)

1. Obtain the weekly demand statistics

$D_w$  = mean weekly demand

$\sigma_w$  = std. dev. of the weekly demand

(from data obtained in the store)

2. Calculate the demand statistics for the supplier lead time,  $L$

$(D_L)_m \triangleq$  mean value of  $D_L$

$\sigma_L \triangleq$  std. dev. of  $D_L$

13.5

Fact (from Prob. theory) : Mean values are additive  
(Assumption :  $D_w$  is the same for each week)

$$(D_L)_m = \underbrace{D_w + D_w + \dots + D_w}_{L \text{ weeks} \equiv L \text{ terms}}$$

$$\boxed{(D_L)_m = L D_w} \longrightarrow (1)$$

Fact (from Prob. theory) = std. dev. is not additive, but  
variance = (std. dev.)<sup>2</sup> is additive

If  $\sigma_w$  is the same for each week,

$$\text{Variance of } D_L = (\sigma_L)^2 = \underbrace{\sigma_w^2 + \sigma_w^2 + \dots + \sigma_w^2}_{L \text{ weeks} \equiv L \text{ terms}}$$

$$\Rightarrow \sigma_L^2 = L \sigma_w^2$$

$$\boxed{\sigma_L = \sqrt{L} \sigma_w} \longrightarrow (2)$$



### 3. Basic relationships for safety stock 13.6

(See diagram in L#12)

$$ROP = (D_L)_m + SS \quad \rightarrow 3(a)$$

$$ROP \stackrel{(1)}{=} L D_w + SS \quad \rightarrow 3(b)$$

$$SS \stackrel{(3(a))}{=} ROP - (D_L)_m \quad \rightarrow 3(c)$$

$$SS \stackrel{(3(b))}{=} ROP - L D_w \quad \rightarrow 3(d)$$

### 4. Product Availability condition (PAC)

During the supplier lead-time,  $L$ ,  
if the actual demand,  $D_L \leq$  available supply,  $ROP$ ,  
then the product is available to all potential  
customers during  $L$ :

$$PAC : D_L \leq ROP \text{ (during } L \text{)}$$

$$\text{If, } \frac{D_L - (D_L)_m}{\sigma_L} \leq \frac{ROP - (D_L)_m}{\sigma_L}, \rightarrow (4)$$

then the product is available to all  
customers during the lead-time,  $L$ .

13.7

Define a variable,  $z \triangleq \frac{D_L - (D_L)_m}{\sigma_L} \rightarrow (5)$

Let  $z' = z \Big|_{D_L = ROP} \stackrel{(5)}{=} \frac{ROP - (D_L)_m}{\sigma_L} \rightarrow (6)$

Product availability condition, eqn. (4), becomes:

If  $z \leq z' \rightarrow (7)$   
(4,5,6)

then the product is available to all potential customers during the lead-time,  $L$ .

Define the cycle service level, CSL, as follows:

$CSL = \text{Probability} (z \leq z') \rightarrow (8)$

[Informally, CSL is the probability of the item being in stock during the lead-time,  $L$ ]

## 5. Interlude into Probability theory

Assumption : Weekly demand can be modeled using a Gaussian or Normal probability density function

$f(z) \triangleq$  probability density function

$F(z) \triangleq$  probability distribution function  
or cumulative probability distribution function

Rule for  
calculating  
Probabilities

$$\text{Probability } (z \leq z') \equiv \text{Prob. } (-\infty < z \leq z') = F(z=z') \rightarrow (9)$$

In words : To obtain the probability of  $z \leq z'$ , simply compute  $F(z)$  at  $z = z'$

Examples : Use statistical tables (on Canvas)

(i)  $\text{Prob. } (z \leq z' = 1.65) \stackrel{(9)}{=} F(z = z' = 1.65) = 0.9505$

(ii)  $\text{Prob. } (z \leq z' = 1.75) \stackrel{(9)}{=} F(z' = 1.75) = \underbrace{\quad}_{\text{(for you)}} ?$